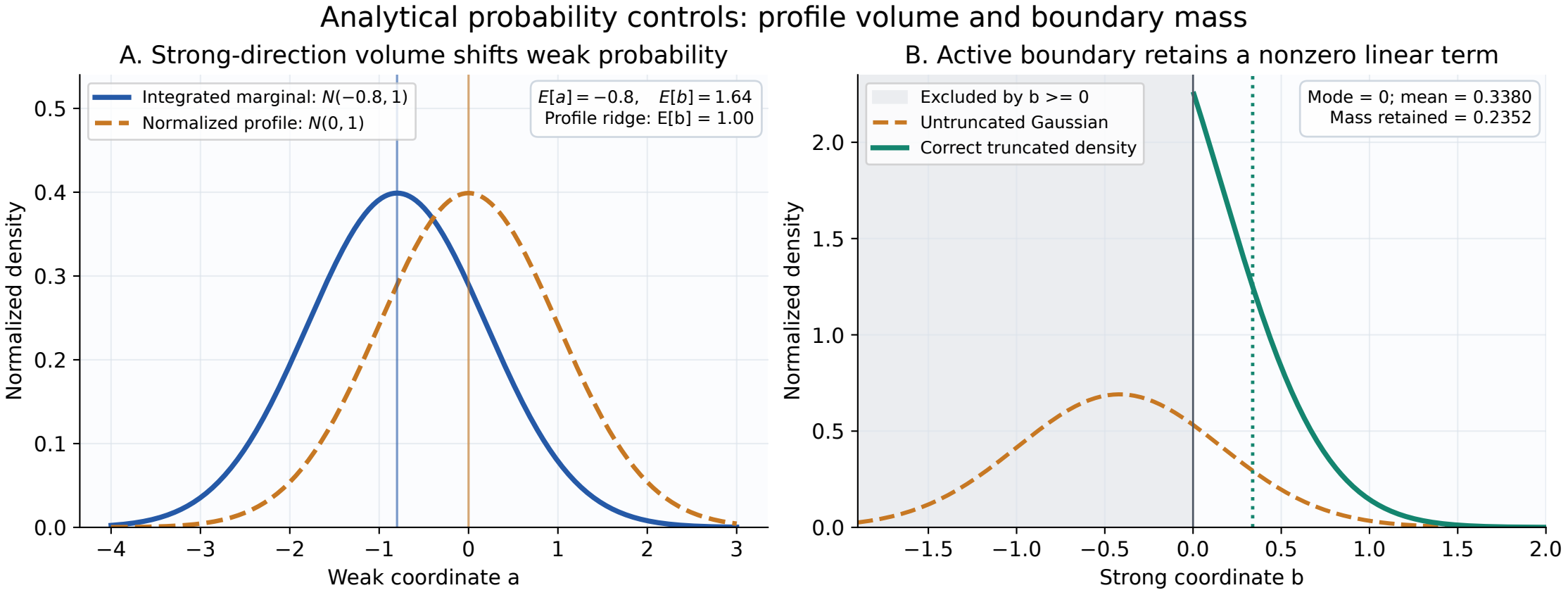
Analytical probability controls: profile volume and boundary mass
A. Strong-direction volume shifts weak probability
Integrated marginal: $N(-0.8, 1)$
Normalized profile: $N(0, 1)$
$E[a] = -0.8, \quad E[b] = 1.64$
Profile ridge: E[b] = 1.00
Normalized density
Weak coordinate a
B. Active boundary retains a nonzero linear term
Excluded by b >= 0
Untruncated Gaussian
Correct truncated density
Mode = 0; mean = 0.3380
Mass retained = 0.2352
Normalized density
Strong coordinate b